The Man Who Understood Entropy:

Correcting the Scientific Record on the Biggest Open Question in Science

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In memory of Gian-Carlo Rota (1932–1999), who understood entropy.

Abstract

Gian-Carlo Rota proved that entropy is unique: every function on finite probability distributions satisfying a short list of natural properties—continuity, conditional additivity, maximality at uniformity, and invariance under zero-probability outcomes—must be a positive scalar multiple of Shannon entropy $H(p_1, \ldots, p_n) = -C \sum p_i \log p_i$. This result, taught at MIT for decades but never formally published or peer-reviewed, settles the question "What is entropy?" that has confounded physicists and information theorists since Clausius coined the term in 1865. We present the fully modernized proof of Rota's Entropy Theorem (RET), machine-verified in Lean 4 with no `sorry` and no custom axioms. In the Lean formalization, Rota's original five properties become seven explicit axioms, all of which are proven as theorems for the concrete Shannon entropy function. The capstone theorem, `RET_All_Entropy_Is_Scaled_Shannon_Entropy`, establishes that every entropy function satisfying these properties is $C \times H_{\text{Shannon}}$ for some $C > 0$. Entropy is not disorder, not surprise, not tendency toward chaos. Entropy is the *exact count of information*—the number of bits required to encode a system's state optimally—and Rota proved this is the only possible such measure.

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1 Introduction: The Question Nobody Could Answer

When Claude Shannon introduced his theory of communication in 1948, he needed a name for his central quantity, $H = -\sum_i p_i \log p_i$. The legend, recounted by Myron Tribus, has John von Neumann advising Shannon:

> "You should call it entropy, for two reasons. In the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, *no one really knows what entropy is*, so in a debate you will always have the advantage."

Von Neumann's quip was not mere humor. By 1948, "entropy" had already accumulated a century of conflicting interpretations:

- **Thermodynamics** (Clausius, 1865): a quantity related to heat transfer and irreversibility, $dS = \delta Q/T$.
- **Statistical mechanics** (Boltzmann, 1877; Gibbs, 1902): a measure of the number of microstates consistent with a macrostate, $S = k_B \ln \Omega$.
- **Information theory** (Shannon, 1948): $H = -\sum p_i \log_2 p_i$, the minimum number of bits per symbol for lossless compression.
- **Quantum mechanics** (von Neumann, 1932): $S = -\mathrm{Tr}(\rho \ln \rho)$, the entropy of a density matrix.

Each community adopted its own notation, its own base of logarithm, its own physical constant of proportionality, and its own informal explanation. Entropy has been variously described as "disorder," "surprise," "uncertainty," "missing information," and "tendency toward chaos." None of these is a precise mathematical definition. All of them are metaphors layered atop one another for 160 years.

The persistent confusion is not merely pedagogical. It reflects a genuine open question: *Why is the same formula $-\sum p_i \log p_i$ the right measure of information, of thermodynamic irreversibility, and of quantum entanglement?* Is this a cosmic coincidence, or is there a mathematical reason?

Gian-Carlo Rota answered this question definitively. He proved that the formula is not a choice—it is a mathematical necessity. Any function satisfying a handful of self-evident properties *must* be Shannon entropy, up to a positive multiplicative constant. All the entropies listed above—Shannon's, Boltzmann's, Gibbs's, von Neumann's—are the same function wearing different units.

Rota's proof was taught at MIT for decades, embedded in a ∼400-page unpublished manuscript co-authored with Kenneth Baclawski and Avner Billis. It was never formally published. It never went through peer review. And so it remained largely unknown to the broader scientific community, even as physicists, computer scientists, and engineers continued to debate what entropy "really means."

This paper aims to correct that record.

2 The Man: Gian-Carlo Rota (1932–1999)

Gian-Carlo Rota was born in Vigevano, Italy, in 1932. His family fled Mussolini's Italy in 1946, emigrating first to Ecuador and then to the United States, where Rota enrolled at Princeton. He completed his Ph.D. at Yale in 1956 under the direction of Jacob T. Schwartz, then joined the faculty at MIT, where he remained for the rest of his career.

Rota is best known as the founder of modern algebraic combinatorics. His 1964 paper "On the Foundations of Combinatorial Theory I: Theory of Möbius Functions" transformed a collection of ad hoc counting tricks into a rigorous algebraic theory, launching an entirely new field. He won the Steele Prize in 1988 and was elected to the National Academy of Sciences in 1982.

Less well known, but equally significant, is Rota's place in the intellectual lineage from von Neumann. Von Neumann was best friends and, at the Manhattan Project, intellectually inseparable from Stanisław Ulam; Ulam, in turn, profoundly influenced Rota during Rota's time at Los Alamos. The chain von Neumann $\to$ Ulam $\to$ Rota represents three generations of mathematicians for whom probability, information, and computation were not separate disciplines but facets of a single structure.

For over two decades, Rota taught **18.313**—"Introduction to Probability"—at MIT. The course notes, co-authored with Baclawski and Billis, grew into a massive manuscript. Chapter 7 of the 1979 edition (Chapter 8 of the 1992 revision) contains the entropy uniqueness theorem. Rota himself warned the reader:

> "The proof is rather technical, so we suggest omitting it on the first reading."
> —Rota, Baclawski, Billis (*Introduction to Probability and Random Processes*, Ch. 8)

The manuscript was never published by a press, never submitted to a journal, and never subjected to formal peer review. Copies circulated among Rota's students and colleagues, but the entropy theorem remained hidden in plain sight—taught to generations of MIT students, yet invisible to the broader community debating the meaning of entropy.

Rota died on April 18, 1999, at the age of 66. His entropy theorem deserves to be part of the scientific record. This paper, and its accompanying machine-verified Lean 4 formalization, aims to put it there.

3 Rota's Five Properties of Entropy

Let Π_n denote the set of *probability distributions* on n outcomes: vectors $(p_1, \ldots, p_n)$ with $p_i \geq 0$ and $\sum_{i=1}^{n} p_i = 1$. Let $\mathcal{H}$ be a function that assigns a non-negative real number to every finite probability distribution. Rota identified five properties that any reasonable entropy function should satisfy.

Axiom 3.1 (Defined on probability distributions)**.** $\mathcal{H}$ is defined on Π_n for every $n \geq 1$. That is, $\mathcal{H}$ takes as input a finite probability distribution summing to 1 and returns a non-negative real number.

Axiom 3.2 (Zero invariance)**.** Adding outcomes with probability 0 does not change the entropy:

$$\mathcal{H}(p_1, \ldots, p_n, 0) = \mathcal{H}(p_1, \ldots, p_n).$$

Axiom 3.3 (Continuity)**.** $\mathcal{H}$ is a continuous function of its arguments. That is, for any distribution $\mathbf{p} \in \Pi_n$ and any $\varepsilon > 0$, there exists $\delta > 0$ such that if $|q_i - p_i| < \delta$ for all i, then $|\mathcal{H}(\mathbf{q}) - \mathcal{H}(\mathbf{p})| < \varepsilon$.

Axiom 3.4 (Conditional additivity)**.** If a system is decomposed into a prior distribution σ and conditional distributions π_i for each outcome i of σ, then:

$$\mathcal{H}(\text{joint}) = \mathcal{H}(\sigma) + \sum_i \sigma_i \cdot \mathcal{H}(\pi_i).$$

This is the chain rule for entropy: the information in the whole equals the information in the first stage plus the expected information in the second.

Axiom 3.5 (Maximum at uniform distribution)**.** Among all distributions on n outcomes, $\mathcal{H}$ is maximized by the uniform distribution:

$$\mathcal{H}(p_1, \ldots, p_n) \leq \mathcal{H}\left(\tfrac{1}{n}, \ldots, \tfrac{1}{n}\right).$$

Remark 3.6. Rota's Axioms 3.1–3.5 correspond to those in the 1979 edition (Chapter 7) and the 1992 revision (Chapter 8) of the Rota–Baclawski manuscript. The statement of conditional additivity (Axiom 3.4) is sometimes written as $H(\pi \mid \sigma) = H(\pi) - H(\sigma)$ in Rota's notation, which is equivalent to the chain rule form above. The key insight is that these five properties *uniquely determine* $\mathcal{H}$ up to a positive multiplicative constant.

4 The Proof: From Axioms to Uniqueness

Rota's proof proceeds in six stages. We outline each, noting the corresponding Lean theorem names in the formalization.

4.1 Stage 1: Entropy of certainty is zero

Define $f(n) = \mathcal{H}(1/n, \ldots, 1/n)$—the entropy of the uniform distribution on n outcomes. From Axioms 3.1 and 3.2, we immediately obtain:

$$f(1) = 0.$$

A distribution concentrated on a single outcome carries no information.

Lean: `f0_1_eq_0`

4.2 Stage 2: Monotonicity

From Axiom 3.5 (maximum at uniform) and Axiom 3.2 (zero invariance), we deduce that $f(n)$ is monotone non-decreasing:

$$n \leq m \implies f(n) \leq f(m).$$

More outcomes means at least as much entropy.

Lean: `f0_mono`

4.3 Stage 3: The Cauchy functional equation

This is the crucial algebraic step. Consider the product of two uniform distributions: one on m outcomes and one on n outcomes. The joint distribution is uniform on $m \cdot n$ outcomes. By the conditional additivity axiom (Axiom 3.4):

$$f(m \cdot n) = f(m) + f(n).$$

This is the Cauchy functional equation for the function f restricted to the positive integers.

Lean: `f0_mul_eq_add_f0`

4.4 Stage 4: The logarithmic trapping argument

This is the heart of Rota's proof and its most technically demanding passage. Fix $b \geq 2$ (we take $b = 2$) and consider any $n \geq 2$. For any positive integer m, there exists a unique integer k such that:

$$b^k \leq n^m < b^{k+1}.$$

Taking logarithms: $k \leq m \log_b n < k + 1$, so

$$\frac{k}{m} \leq \log_b n < \frac{k+1}{m}.$$

Now apply the monotonicity and power law properties. From $b^k \leq n^m$ and monotonicity of f:

$$k \cdot f(b) \leq m \cdot f(n).$$

From $n^m < b^{k+1}$ and monotonicity:

$$m \cdot f(n) \leq (k+1) \cdot f(b).$$

If $f(b) > 0$ (i.e., $\mathcal{H}$ is non-trivial), dividing through yields:

$$\frac{k}{m} \leq \frac{f(n)}{f(b)} \leq \frac{k+1}{m}.$$

Both $f(n)/f(b)$ and $\log_b n$ are trapped in the same interval of width $1/m$. Since m is arbitrary, letting $m \to \infty$ forces:

$$\frac{f(n)}{f(b)} = \log_b n \qquad \Longrightarrow \qquad f(n) = f(b) \cdot \log_b n = C \cdot \log n,$$

where $C = f(b)/\log b$ is a positive constant independent of n.

Lean: `logarithmic_trapping`, `uniformEntropy_power_law`

4.5 Stage 5: Extension to rational probabilities

For a distribution with rational probabilities $p_i = a_i/N$ where $\sum a_i = N$, the conditional additivity axiom yields:

$$\mathcal{H}(p_1, \ldots, p_n) = f(N) - \sum_i p_i \cdot f(a_i) = C \left(\log N - \sum_i p_i \log a_i \right) = C \sum_i p_i \log \frac{1}{p_i}.$$

This is precisely C times Shannon entropy.

Lean: `RUE_rational_case`

4.6 Stage 6: Extension to real probabilities

By the continuity axiom (Axiom 3.3), the result for rational probabilities extends to all real-valued probabilities by density of $\mathbb{Q}$ in $\mathbb{R}$. We conclude:

Theorem 4.1 (Rota's Entropy Theorem—Uniqueness)**.** *Let $\mathcal{H}$ be a function on finite probability distributions satisfying Axioms 3.1–3.5. Then there exists a constant $C \geq 0$ such that for all $(p_1, \ldots, p_n) \in \Pi_n$:*

$$\mathcal{H}(p_1, \ldots, p_n) = C \sum_{i=1}^{n} p_i \log \frac{1}{p_i}.$$

If $\mathcal{H}$ is non-trivial (i.e., not identically zero), then $C > 0$.

Lean: `RotaUniformTheorem`, `RotaUniformTheorem_formula_with_C_constant`

5 The Modernization: From Pen-and-Paper to Machine-Verified Proof

The formalization was carried out in Lean 4 using the mathlib4 library. It lives in the `EGPT/Entropy/` module and spans three files: `Common.lean` (axiom definitions), `RET.lean` (the abstract uniqueness proof), and `H.lean` (concrete verification that Shannon entropy satisfies all axioms). A fourth file, `NumberTheory/Analysis.lean`, provides the capstone theorem.

5.1 From five properties to seven axioms

Rota's five pen-and-paper properties translate into seven explicit Lean axioms bundled in the structure `HasRotaEntropyProperties`:

#	Axiom (Lean name)	Origin
1	`IsEntropyNormalized`	Implicit in Rota (from Ax. 3.1+3.2)
2	`IsEntropySymmetric`	Implicit (Rota treats distributions as sets)
3	`IsEntropyZeroOnEmptyDomain`	Lean type-system edge case (Fin 0)
4	`IsEntropyZeroInvariance`	Axiom 3.2
5	`IsEntropyContinuous`	Axiom 3.3
6	`IsEntropyCondAddSigma`	Axiom 3.4
7	`IsEntropyMaxUniform`	Axiom 3.5

The first two axioms were implicit in Rota's presentation. *Normalization* ($\mathcal{H}(\{1\}) = 0$) follows trivially on paper from the zero-invariance and probability-set axioms, but making it explicit simplifies the formal proof chain. *Symmetry*—invariance under relabeling of outcomes—was automatic for Rota because he treated distributions as unordered sets; in Lean, where Fin $n \to \mathbb{R}_{\geq 0}$ is an ordered function type, symmetry must be stated explicitly as invariance under composition with an equivalence $\alpha \simeq \beta$. The third axiom, $\mathcal{H}(\emptyset) = 0$, handles the edge case Fin 0 that Rota's framework never considered.

5.2 The abstract uniqueness proof (`RET.lean`)

File `RET.lean` is a faithful rendition of Rota's proof. It assumes an abstract `H_func` satisfying `HasRotaEntropyProperties` and derives the key results listed in Section 4:

- `f0_1_eq_0`: $f(1) = 0$.

- `f0_mono`: f is monotone non-decreasing.
- `f0_mul_eq_add_f0`: $f(mn) = f(m) + f(n)$.
- `uniformEntropy_power_law`: $f(n^k) = k \cdot f(n)$.
- `logarithmic_trapping`: the trapping argument forcing $f(n)/f(b) = \log_b n$.
- `RotaUniformTheorem`: $f(n) = C \cdot \log n$.

The structure `EntropyFunction` bundles an `H_func` together with a proof that it satisfies `HasRotaEntropyProperties`:

```
structure EntropyFunction where
  (H_func : forall {a : Type} [Fintype a],
            (a -> NNReal) -> NNReal)
  (props : HasRotaEntropyProperties H_func)
```

5.3 Proving the axioms as theorems (`H.lean`)

The modernization step that goes beyond Rota's original work is verifying that the concrete Shannon entropy function satisfies all seven axioms. File `H.lean` defines the canonical entropy function `H_canonical_ln`—standard Shannon entropy using the natural logarithm—and proves each axiom as a theorem:

1. `h_canonical_is_normalized`: $H(\{1\}) = 0$.
2. `h_canonical_is_symmetric`: H is invariant under relabeling.
3. `h_canonical_is_zero_on_empty`: $H(\emptyset) = 0$.
4. `h_canonical_is_zero_invariance`: appending a zero-probability outcome does not change H.
5. `h_canonical_is_continuous`: H is continuous in ε-δ sense.
6. `h_canonical_is_cond_add_sigma`: the chain rule holds.
7. `h_canonical_is_max_uniform`: H is maximized by the uniform distribution.

These are bundled into a single witness:

```
noncomputable def TheCanonicalEntropyFunction_Ln :
  EntropyFunction := { ... }
```

This instance certifies that Shannon entropy is a valid `EntropyFunction`—and by the uniqueness theorem, the *only* one (up to scaling).

5.4 The capstone theorem

The culmination appears in `NumberTheory/Analysis.lean`:

```
theorem RET_All_Entropy_Is_Scaled_Shannon_Entropy
  (ef : EntropyFunction) (C : R) (hC_pos : 0 < C) :
  HasRotaEntropyProperties
    (fun p => (C * (ef.H_func p : R)).toNNReal)
```

This says: given *any* entropy function `ef` satisfying the seven axioms, and *any* positive constant C, the scaled function $C \cdot H$ again satisfies all seven axioms. Combined with the uniqueness result from `RET.lean`, this establishes that the space of entropy functions is the one-dimensional ray $\{C \cdot H_{\text{Shannon}} : C > 0\}$.

The entire chain—from axiom definitions through the abstract uniqueness proof to the concrete verification and the capstone—is `sorry`-free and uses no custom axioms.

As a final "unit test," the formalization proves:

```
theorem entropy_of_fair_coin_is_one_bit :
  H_canonical_log2 (canonicalUniformDist 2 ...) = 1
```

A fair coin flip has exactly one bit of entropy. The engine is calibrated.

6 What Entropy Actually Is

With Rota's theorem in hand, we can state plainly what entropy is and what it is not.

Entropy is not disorder. The "disorder" metaphor, ubiquitous in textbooks, has no mathematical content. It confuses a human judgment ("this looks messy") with a precisely defined quantity. A shuffled deck of cards and a sorted deck occupy the same state space; neither is more or less "disordered" in any formal sense.

Entropy is not surprise. The "surprise" interpretation—common in information theory pedagogy—captures intuition about rare events carrying more information. But surprise is a property of a single outcome; entropy is a property of an entire distribution.

Entropy is not tendency toward chaos. This conflates a state function (entropy) with a dynamical law (the second law of thermodynamics). Entropy does not "tend" anywhere. The second law says the entropy of an isolated system does not decrease; entropy itself is just a number attached to a probability distribution.

Entropy is the exact count of information. RET proves that the *unique* function satisfying Axioms 3.1–3.5 is:

$$\mathcal{H}(p_1, \ldots, p_n) = C \sum_{i=1}^{n} p_i \log \frac{1}{p_i},$$

for some constant $C > 0$. When $C = 1$ and the logarithm is base 2, this is the number of bits required to encode a random variable optimally. When $C = k_B$ and the logarithm is natural, this is Boltzmann's entropy. The formula is the same; only the units change.

Entropy counts information. The logarithm is not an arbitrary choice—it is a mathematical *necessity*, forced by the monotonicity and multiplicative-to-additive property of f. Any monotone function satisfying $f(mn) = f(m) + f(n)$ is a logarithm; the trapping argument of Section 4 makes this rigorous.

The double-entry accounting metaphor. If truth is the record of what actually happened in the past, then entropy is its ledger. Every physical process that resolves uncertainty creates information; entropy is the quantity that ensures the books balance. Rota's theorem is the double-entry accounting system: it guarantees that there is exactly one way to keep the ledger, up to a choice of currency (the constant C). You may measure in bits, nats, or joules per kelvin, but the ledger is the same.

All entropies are one. Shannon entropy, Boltzmann entropy, Gibbs entropy, von Neumann entropy, Rényi entropy of order 1—all are the same function $C \sum p_i \log(1/p_i)$ with different values of C and different bases of logarithm. RET unifies them.

7 Implications

7.1 Einstein's question answered

Einstein spent decades troubled by the tension between the continuous fields of general relativity and the discrete quanta of quantum mechanics. If entropy is the unique measure of information, and information is the same whether encoded in continuous or discrete form, then the apparent tension dissolves.

The Lean formalization proves that the continuous Shannon entropy $H_{\text{continuous}}$ and the discrete Shannon entropy H_{discrete} are related by $H_{\text{continuous}} = C \times H_{\text{discrete}}$ for a universal constant C. Continuous and discrete descriptions are not different physics—they are different encodings of the same information.

7.2 The AI energy question

If entropy equals computation, and computation is counting, then the optimal encoding of computation is exact arithmetic—integer operations (IOPs) rather than floating-point operations (FLOPs). Floating-point arithmetic introduces rounding error at every step; error accumulation then demands larger word sizes, more memory, and more energy to compensate.

The EGPT integer-only math library (EGPTMath) demonstrates this principle: exact arithmetic using prime-power factorized number representations replaces every FLOP with an IOP. The Shannon-coding interpretation of entropy guarantees that this encoding is optimal—no bits are wasted, and no error accumulates. The implication for AI hardware is clear: if computation can be made exact, the energy cost of error correction vanishes.

7.3 Physics unification via statistical distributions

The Lean formalization extends beyond the abstract uniqueness theorem. The accompanying `Physics/` module proves that all three canonical statistical distributions—Bose–Einstein, Fermi–Dirac, and Maxwell–Boltzmann—satisfy $H = C \times H_{\text{Shannon}}$ over $\mathbb{R}$. This means that the entropy of *every* physical system governed by quantum or classical statistics is a scaled Shannon entropy. There is no separate "physical" entropy and "information" entropy; there is only entropy.

8 Conclusion: Credit Where It Is Due

Gian-Carlo Rota solved the entropy question decades ago. His proof established, with mathematical certainty, that entropy is unique: every function satisfying a small number of natural axioms is a constant multiple of Shannon entropy. The logarithm is not a convention but a theorem. The proliferating definitions of entropy across physics, information theory, and computer science are not competing theories but different units of the same measure.

The result was available but hidden. Buried in an unpublished manuscript, taught to MIT students but never submitted for publication, Rota's entropy theorem missed its moment. The scientific community continued debating what entropy means while the answer sat in a filing cabinet in Cambridge, Massachusetts.

This paper and its accompanying Lean 4 formalization aim to correct the record. The proof is now machine-verified, permanent, and public. Every axiom is a proven theorem. Every step from axioms to uniqueness has been checked by a computer. There are no `sorry`s, no custom axioms, no gaps.

The intellectual debt runs deep. Von Neumann, who told Shannon that "no one really knows what entropy is," was one mind at the Manhattan project with Ulam, who influenced Rota, who proved that the answer was always implicit in the question. Entropy is the count of information. It always was.

It is time the world knew.

Acknowledgments

The author thanks the Lean community and the mathlib4 contributors whose library made the formalization possible. The original ideas belong to Gian-Carlo Rota, Kenneth Baclawski, and Avner Billis. This paper merely formalizes and publicizes what they created.

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A Summary of Lean Theorem Names

For reference, the complete chain of formal results:

Lean identifier	**Statement**
Abstract uniqueness (`RET.lean`)	
`f0_1_eq_0`	$f(1) = 0$
`f0_mono`	f is monotone
`f0_mul_eq_add_f0`	$f(mn) = f(m) + f(n)$
`uniformEntropy_power_law`	$f(n^k) = k \cdot f(n)$
`logarithmic_trapping`	Trapping argument
`RotaUniformTheorem`	$\exists C,\ f(n) = C \log n$
Concrete axiom proofs (`H.lean`)	
`h_canonical_is_normalized`	$H(\{1\}) = 0$
`h_canonical_is_symmetric`	Relabeling invariance
`h_canonical_is_zero_on_empty`	$H(\emptyset) = 0$
`h_canonical_is_zero_invariance`	Zero-probability invariance
`h_canonical_is_continuous`	ε-δ continuity
`h_canonical_is_cond_add_sigma`	Chain rule
`h_canonical_is_max_uniform`	Maximum at uniform
Bundled witness (`H.lean`)	
`TheCanonicalEntropyFunction_Ln`	Shannon entropy is an `EntropyFunction`
Capstone (`Analysis.lean`)	
`RET_All_Entropy_Is_Scaled_Shannon_Entropy`	$\forall C > 0$: $C \cdot H$ satisfies all axioms
Calibration (`H.lean` / `Analysis.lean`)	
`entropy_of_fair_coin_is_one_bit`	$H_{\log_2}(\text{fair coin}) = 1$ bit